



**THIS IS A PRE-SOLICITATION NOTICE. THIS IS NOT A REQUEST FOR QUOTES.** The United States Department of Agriculture (USDA), Agriculture Research Service (ARS) is posting this pre-solicitation notice in accordance with FAR 5.101.

Point of Contact:

Frank Palmer, Contracting Officer

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The proposed contract action will be unrestricted. The North American Industry Classification System (NAICS) 334516 – Analytical Laboratory Instrument Manufacturing, which contains a size standard of 1,000 employees is applicable to this acquisition.

Description:

This requirement is for an Oxygen Permeation (OX-TRAN 2/22 H) and Water Vapor Permeation Analyzer (AQUATRAN 3/34 H), or a functionally equivalent single-cell system, meeting the following high-barrier technical specifications.

**Oxygen Transmission Rate (OTR) Testing:**

- **Detection Range:** 0.05 to 200 cc/(m<sup>2</sup>·day).
- **Repeatability:** 0.02 cc/(m<sup>2</sup>·day) or 1% whichever is higher.
- **Sensor:** Must utilize a coulometric sensor (e.g., COULOX) that requires no calibration and complies with ASTM D3985.
- **Compliance:** Fully compliant with ASTM D3985, F1307, and F1927, as well as ISO, JIS, DIN, and GB/T international standards.
- **Included Accessories:** Starter kit with certified film, cutting kit, template, Apiezon grease, and a specialized **two-stage stainless steel diaphragm regulator set** for carrier gas (N<sub>2</sub>/H<sub>2</sub>) and oxygen.

**Water Vapor Transmission Rate (WVTR) Testing:**

- **Detection Range:** 0.05 to 100 g/(m<sup>2</sup>·day).
- **Repeatability:** 0.05 g/(m<sup>2</sup>·day) or 2%, whichever is greater.
- **Sensor:** Must utilize modulated infrared (IR) technology complying with ASTM F1249.
- **Compliance:** Meets **ASTM F1249**, ISO 15106-3, JIS K-7129, TAPPI T557, and GB/T 26253-2010 standards.
- **Included Accessories:** Starter kit with #1 Yellow and #2 Red certified films, desiccant chamber assembly, and a precision **regulator set for N<sub>2</sub>/He carrier gas**.

**Common System Requirements:**

- The systems to have all the accessories required to **analyse films and finished pouches** with universal adaptor.
- **Temperature Control:** Range of 20 to 40°C ± 0.2°C.
- **RH Control:** OTR range (0%, 30 to 90% ±3%); WVTR range (100% and 50% to 90% ±3%).
- **Software:** Automated testing control with data management capabilities, compatible with Windows 10.

- Includes four precision, two-stage stainless steel diaphragm regulators for both carrier gas and oxygen, with all necessary fittings needed to make accurate pressure adjustments.
- **Calibration:** Instruments include NIST-traceable certification.

#### Warranty Information

- **General:** Full warranty on parts and workmanship for **12 months**.
- **Sensors:** Exclusive **4-year prorated warranty** on both the COULOX and modulated IR sensors.

Quotes for this requirement will be solicited from at least three sources in accordance with FAR 12.201-1 ( c ) (2). This requirement will be awarded based on the lowest price technically acceptable. The anticipated award date is June 2026.

All responsible sources may submit a quotation which will be considered by the agency.

Vendors are required to be active and registered in the System for Award Management at [www.sam.gov](http://www.sam.gov) when submitting a quote and at time of award (FAR 52.204-7). Payment will be made utilizing the Department of the Treasury Invoice Processing Platform at [www.ipp.gov](http://www.ipp.gov). SAM.gov and IPP.gov are FREE, official websites of the United States government.

The Government reserves the right to reject, in whole or in part, any contractor's input resulting from this pre-solicitation notice. No contract will be awarded as a result of this notice. Data submitted in response to this pre-solicitation notice will not be returned. This request is for informational purposes only and the Government will not pay any costs associated with the preparation and submission of a response to this notice.

All interested contractors who can meet the requirements stated above should respond with information describing their interest, capabilities statements, and indicate the small business category for which they qualify. E-mail responses are required. The due date for responses to this pre-solicitation notice is 3:00 pm Mountain Daylight Time on Thursday, May 21, 2026. Please email your responses to [frank.palmer@usda.gov](mailto:frank.palmer@usda.gov).

Electronic files must be in Microsoft Office Word or PDF format. All information must be UNCLASSIFIED material.

Respondents should include as part of their submission:

- a. A synopsis of the company's capabilities to meet the requirements stated above (limit 5 pages).
- b. UEI Number.
- c. Company Name.
- d. Company Address.
- e. Business size as validated via the System for Award Administration (SAM). All offerors must register on the SAM located at <http://www.sam.gov>.
- f. Company points of contact, phone number, and email address.
- g. Past work experience. If you have experience working with other Federal Agencies and/or commercial businesses of similar size and scope, please include agency and/or business name contract number (if applicable), amount and type of contract.

Any company proprietary information must be marked as such.

**THIS IS NOT A REQUEST FOR QUOTE.** Interested firms are reminded that in accordance with FAR 4.12, prospective contractors must complete electronic annual representations and certifications and in conjunction with FAR 4.11, are required to be registered in the System for Award Management (SAM) at the time an offer or quotation is submitted.

**[END OF PRE-SOLICITATION NOTICE]**